Read this first

Free test version | 12 November 2026

About this guide

The following 13 pages walk you through setup and use step by step. The screenshots come from the 7 November 2026 version; menus and workflows are unchanged.

Download and installation

Download: github.com/JoshuaGerke/RobertsCrewChiefAdvanced under “Releases” (aktuell). Extract the entire ZIP into a new folder, for example C:\Simracing\RRCA, and start CrewChiefV4.exe there. RRCA uses its own folders for profiles and sounds; your normal Crew Chief stays untouched. Do not run both for the same sim at once, and do not install a Crew Chief update over the RRCA folder.

Crew Chief voice commands

RRCA does not ship Microsoft’s Microsoft.Speech.dll. Crew Chief’s classic English voice commands therefore use the Windows speech recogniser. Install English with speech recognition in Windows under Settings → Time & language → Language & region. Ben, Mik and Sara do not need this: they listen via their own radio button and cloud speech recognition.

Costs, keys and bug reports

RRCA is free. AI and voices need your own OpenRouter and ElevenLabs accounts; using them may cost money. Your keys are stored encrypted in your Windows profile and are never shipped with the package. Please report bugs with the version (VERSION.txt in the package), sim, track, time and the call you heard in the RRCA thread on forum.virtualracing.org.

Your AI crew for Crew Chief

What it is, what you need and how to get from extraction to your first radio call.

RRCA adds Ben, Mik and Sara to Crew Chief. The crew evaluates available simulation data, prioritises confirmed events and combines local calls with AI conversations and your chosen, natural-sounding voices.

Crew Chief remains the foundation

The classic Crew Chief window and RRCA belong to the same extended application. The old window can remain minimised in the background. Do not start a second Crew Chief alongside it.

The three roles

Ben: strategy and race observation.
Mik: car, tyres and technical questions.
Sara: immediate side warnings from the local spotter processing.

Before your first test

Alpha software can give incorrect, late or missing calls. This version has not yet passed a live driving acceptance test. Race radio and Sara clips are currently German; the interface offers DE/EN/ES.



Free, non-commercial hobby project based on Crew Chief V4 (Britton IT Ltd, MIT). Thank you to its developers. Additional personal permission from the developers has not been documented. External AI and voice services may cost money.

1 | What you need

Users do not need development tools, Python or programming knowledge.

1. A 64-bit Windows PC; this build was checked on Windows 11.
2. .NET Framework 4.8 and Microsoft Edge WebView2 Runtime. Install from official Microsoft sources if missing.
3. A simulation supported by the included Crew Chief adapters, with telemetry / shared memory / UDP enabled or the required plugin installed.
4. Your own OpenRouter account with credit or an eligible plan for AI and speech recognition.
5. Your own ElevenLabs account, authorised voices and enough quota for speech output.
6. A microphone and an easily reached controller or wheel button for push-to-talk.
7. The complete RRCA ZIP, including data and libraries. Copying only the EXE is not enough.
8. Time for preparing voices and a short stationary test. New speech clips may incur API charges.

Not required: a separately running original spotter. RRCA uses the integrated Crew Chief foundation.

2 | Extract safely and start the correct instance

A separate RRCA installation protects your working Crew Chief installation.

1. Back up and close the original. Keep your existing Crew Chief folder as a separate installation. Do not run both Crew Chief instances for the same sim at once.

2. Extract into a new folder, for example C:\Simracing\RRCA. Extract the entire ZIP; do not copy it into the original folder.

3. Run CrewChiefV4.exe from the RRCA folder. RRCA opens in front; the classic Crew Chief window is underneath or minimised. It remains part of the same running application.

4. Set up the base sounds. Use the Crew Chief sound manager if no base sounds are found. WAV sound packs and your selected voices are not in the ZIP for rights and privacy reasons.

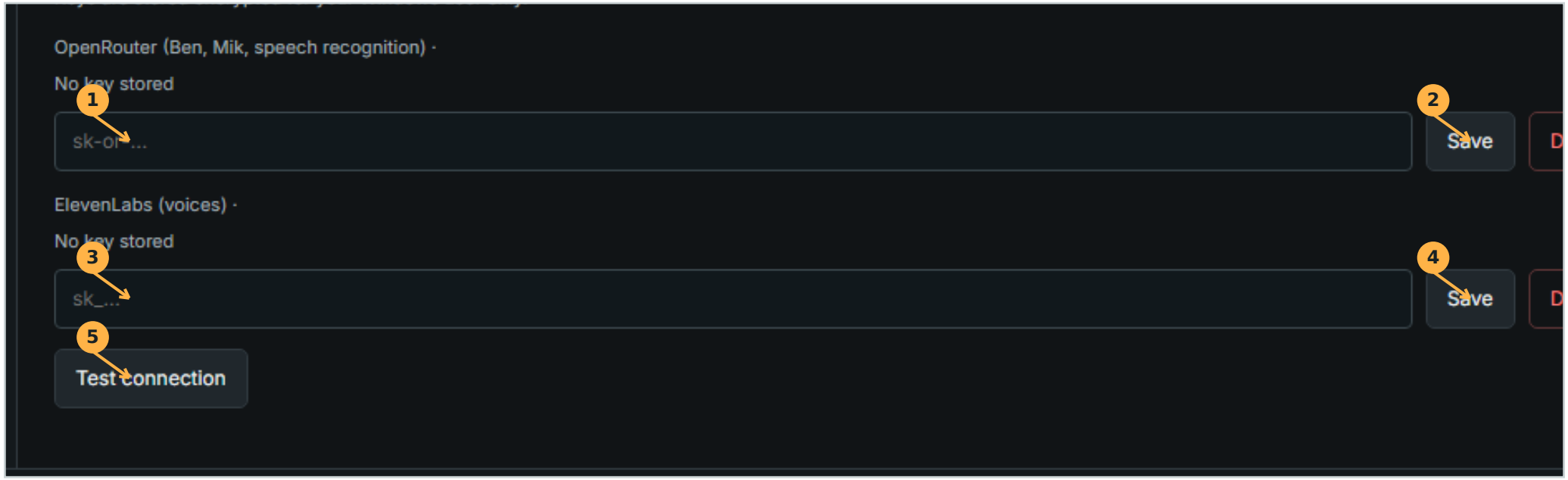
5. Then set up keys, voices and push-to-talk. The following pages show the actual RRCA menus with neutral example values.

Separate folders for RRCA

RRCA keeps its profiles and sounds in its own folders and leaves your normal Crew Chief untouched. Keys and driver memory for the same Windows user are stored outside the ZIP and survive updates.

3 | The two API keys

Team & voices > Services & keys. The screenshots show empty fields, not real credentials.

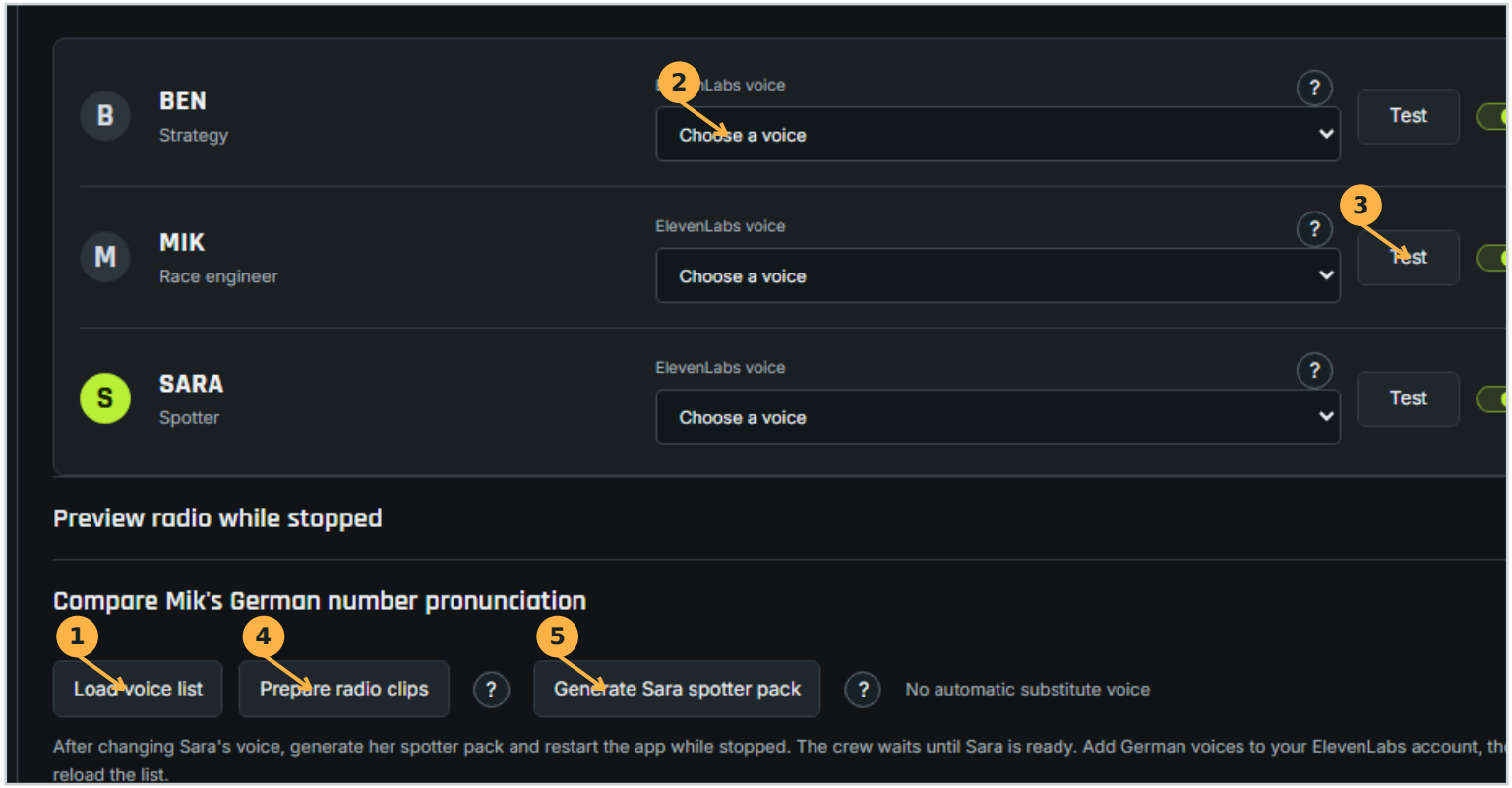


1. OpenRouter: enter your own key for Ben, Mik and speech recognition.
2. Save: store the OpenRouter key encrypted for this Windows user.
3. ElevenLabs: enter your own key for voices and speech output here.
4. Save: save the ElevenLabs key separately. Do not swap the two keys.
5. Test connection: checks the AI connection. Test the microphone and audio separately afterwards.

OpenRouter provides AI access; ElevenLabs is the preferred voice route in this build. Your own provider accounts and their terms apply. Do not create an .env file for RRCA or copy keys into files next to the EXE.

4 | Give Ben, Mik and Sara their voices

Team & voices. Select voices and finish preparation while stopped.

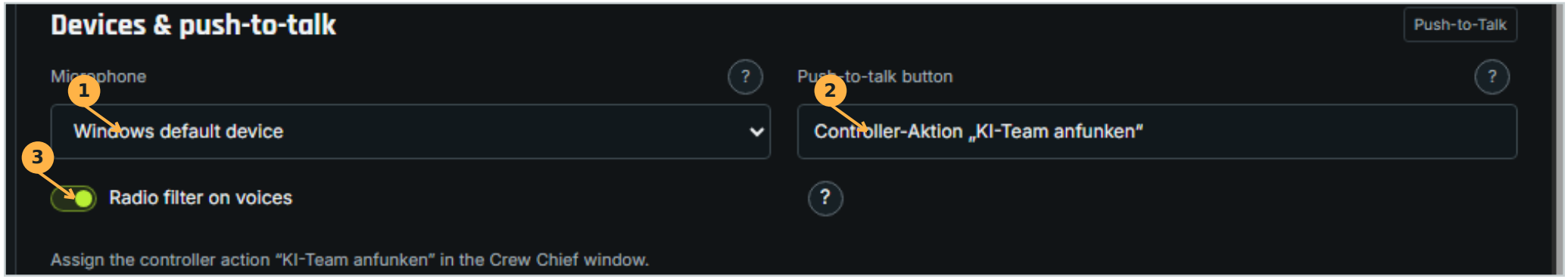


1. Load voice list: shows the voices in your ElevenLabs account.
2. Assign: choose a voice suitable for German for each of Ben, Mik and Sara.
3. Preview: listen to each selected voice. Compare German numbers and names in particular.
4. Prepare radio clips: generate known calls locally; this can consume credit.
5. Generate Sara as spotter: finish her local pack, then fully close and restart the app.

If needed, first add new voices to your ElevenLabs account. Changing a voice, voice model or radio filter may require new clips. A red Sara notice explains what is missing. The app does not add an automatic Windows substitute voice.

5 | Microphone and the familiar Crew Chief button

Team & voices > Devices & push-to-talk. Assign the hardware button in the classic Crew Chief window.

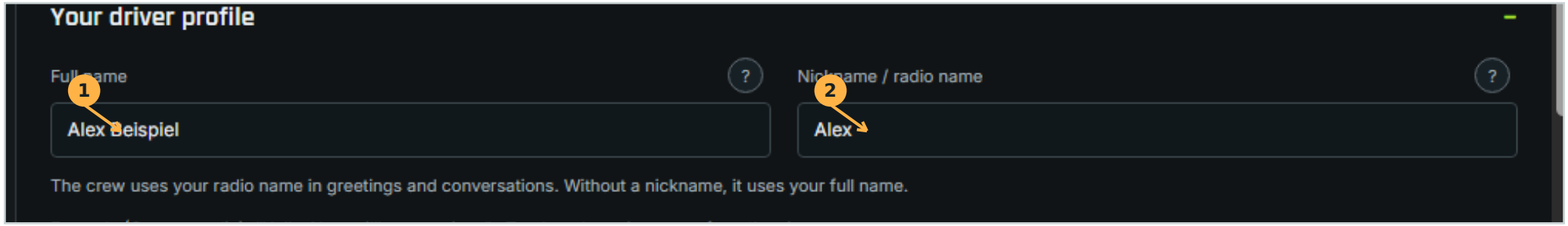


1. Microphone: select the recording device you actually use.
2. Push-to-talk: shows the current action. In the classic window choose the controller, select 'KI-Team anfunken' and click 'Assign'; then press your chosen button.
3. Radio filter: changes the radio sound. Check Sara's pack readiness afterwards.

If the action is missing, open 'Add/Remove Actions'. Then hold the button, speak and release. Return to RRCA after assignment; the classic window may stay minimised. Button assignment is not the entire initial setup: keys, voices, audio and sim telemetry must also work.

6 | How the crew should address you

AI & radio > Your driver profile. 'Alex Beispiel' and 'Alex' are neutral examples.

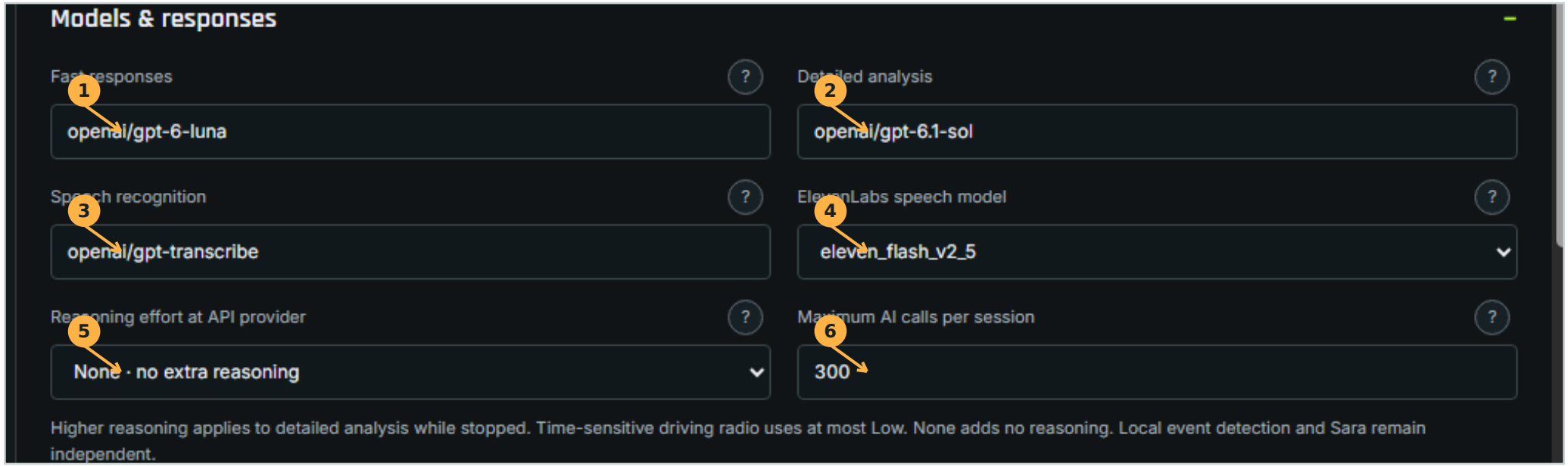


1. Full name: enter your name; this does not create a new user account.
2. Nickname / radio name: enter the short form you want to hear. Without a nickname the full name is used.

Example: 'Hallo Alex, willkommen in ...'. Track and session must actually come from the simulation. Click Apply at the bottom with the crew stopped. A missing greeting when session data is absent or invalid is not evidence that telemetry works.

7 | Which AI does what?

AI & radio > Models & replies. Enter model IDs exactly and check provider quotas.

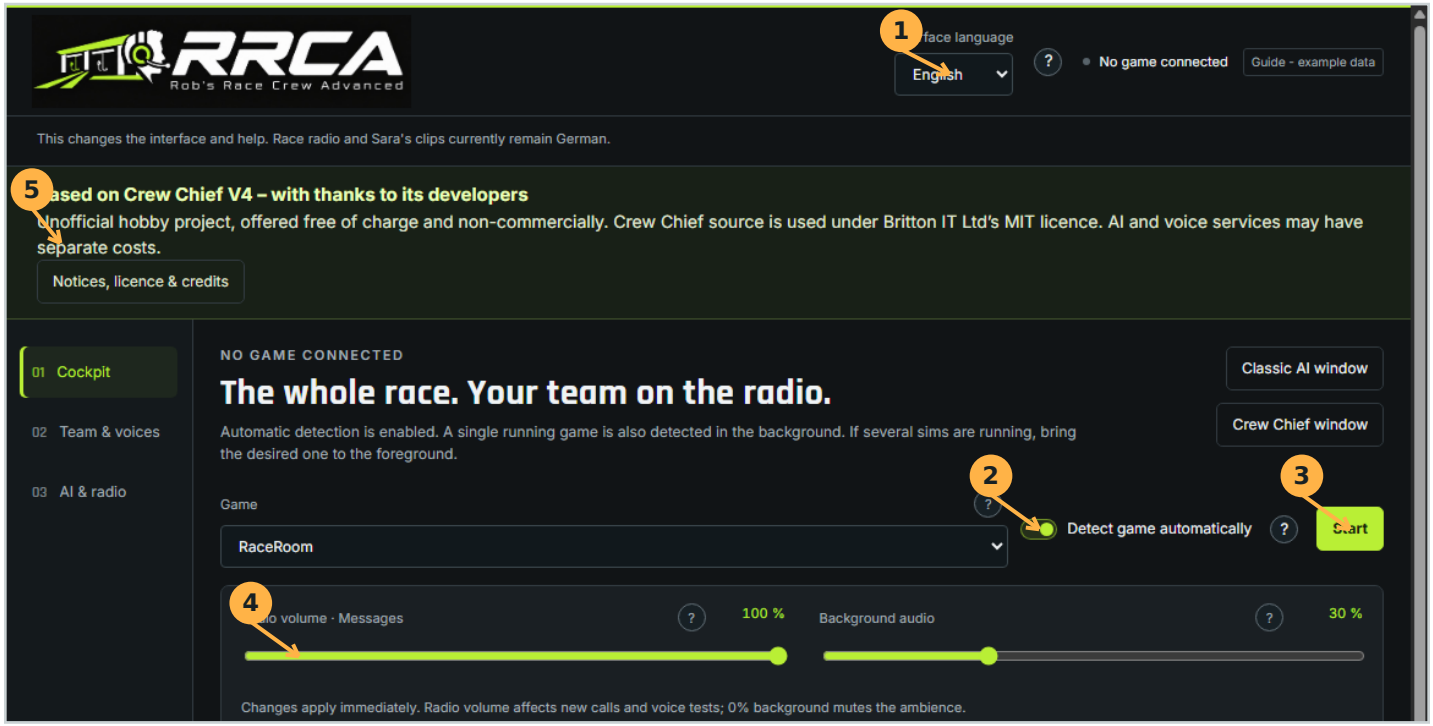


1. Fast replies: default openai/gpt-6-luna for the fast radio route.
2. Detailed analysis: default openai/gpt-6.1-sol for more detailed questions while stopped.
3. Speech recognition: openai/gpt-transcribe is the existing transcription setting; test with a short radio call after setup.
4. ElevenLabs model: start with Flash v2.5 for speed; compare German pronunciation with Multilingual v2 using the same voice.
5. Reasoning: start with None. More reasoning can make stationary analysis slower and more expensive; driving radio uses at most Low.
6. Call limit: initially keep 300. It does not limit every STT/TTS charge.

Both chat IDs were found in the public OpenRouter catalogue on 7 November 2026. Transcription uses a separate audio endpoint and is not verified there as a chat model. Sara and local event detection do not wait for the chat model. New ElevenLabs model names are not automatically supported in this alpha.

8 | Detect the game, start and adjust volume

Cockpit. RRCA and the minimised classic window belong to the same instance.



1. Language: DE, EN or ES for the interface and help. Spoken radio is currently German.
2. Automatic detection: enable for one clearly running supported sim. If several sims are running, bring the desired one to the foreground.
3. Start / Stop: when selecting manually, choose the sim first, then Start. An accessible source can still lack valid driving data while in a menu.
4. Messages: adjusts new radio calls and voice previews. Background is radio ambience; 0% mutes it.
5. Attribution: Crew Chief is the technical foundation. Open licences and notices directly here.

Also check that audio reaches your headset. If needed open the Crew Chief window and check the output device there. Do not change it during a race. For the first stationary test check the microphone, voice preview and a short radio call individually.

9 | Regular radio without talking over the race

AI & radio > Regular radio reports. Event calls remain independent of these ten topics.

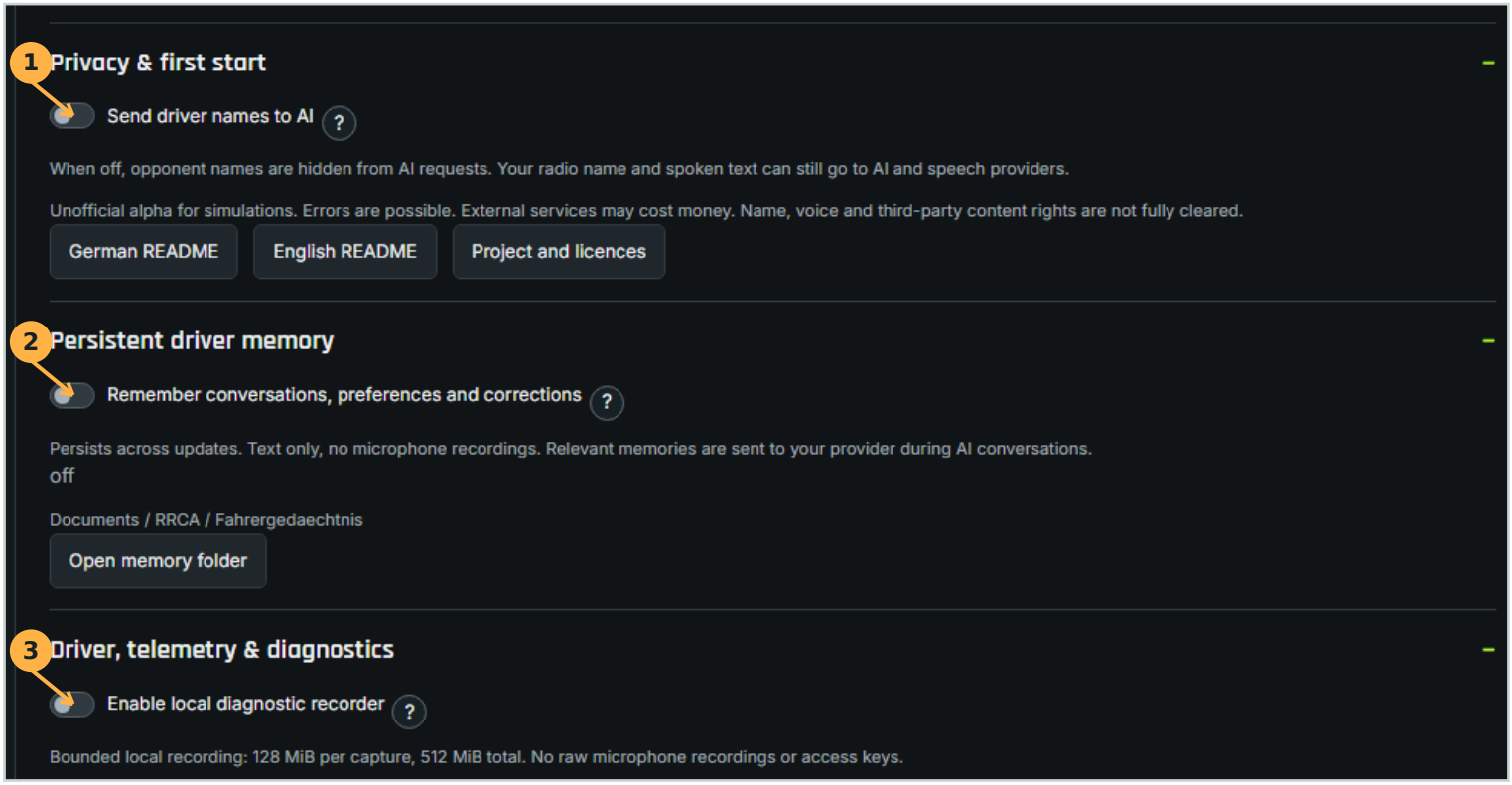


1. Race status: 120 seconds is a sensible starting interval for position and gaps.
2. Weather: start with 300 seconds. Only available values can be reported.
3. Car condition: start with 300 seconds; only supplied technical and damage data.
4. Minus tolerance: an earlier possible time, for example 7 seconds.
5. Plus tolerance: a later possible time, for example 8 seconds.

120 -7/+8 schedules between 113 and 128 seconds. This is not a guaranteed airtime: urgent radio, corner workload, confirmed qualifying push laps and missing values defer reports. Each topic has On/Off. Routine reports are at least 20 seconds apart. Apply model/report settings while stopped.

10 | Memory, names and diagnostics

AI & radio. Expand these sections; your own PC displays the actual storage locations there.

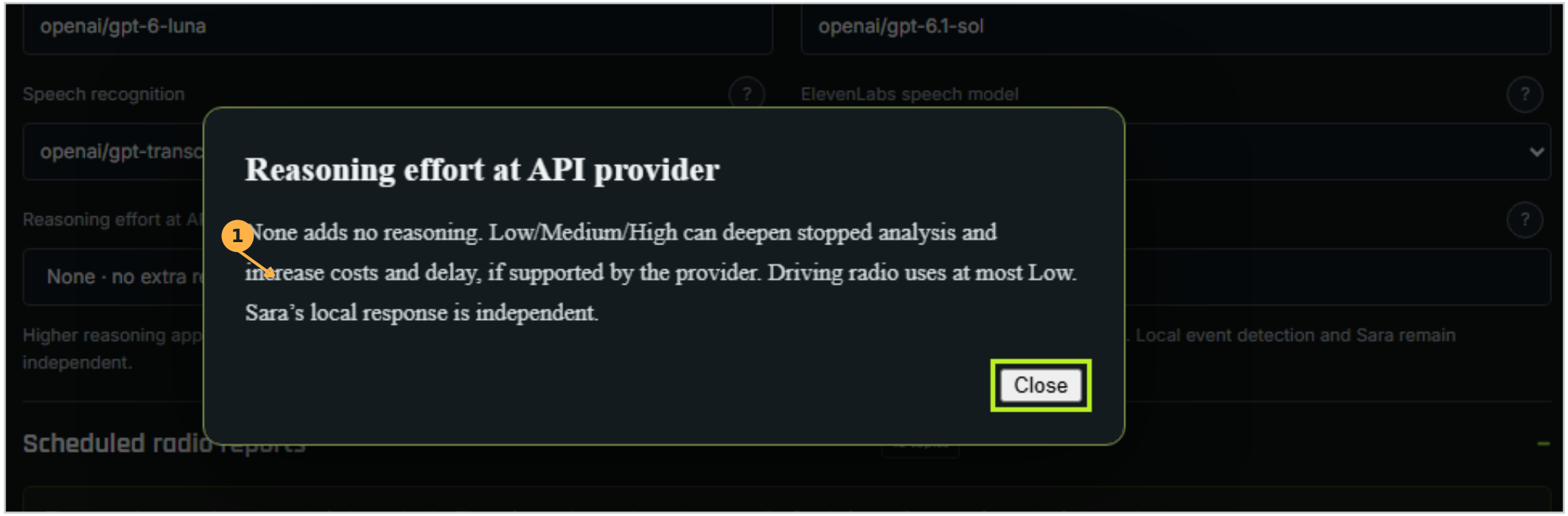


1. Driver names: off in a new profile. Enable to include opponent names in AI requests. Your own radio name and TTS text may still contain names.
2. Memory: stores text, preferences and bounded facts locally. Updates do not remove it. This does not train model weights or change program code.
3. Recorder: enable while stationary before a test. Limited to 128 MiB per recording and 512 MiB total; no complete microphone recordings.

Relevant memories may be sent to the provider during AI conversations. Diagnostic and radio logs can contain names and conversations: review before sharing. Public data paths are in README_EN.md. Finally use Apply at the bottom where required.

11 | Question marks explain the field

A question mark opens a short explanation in your selected interface language.



1 Quick help: read without leaving the app. Closing returns focus to the previous field.

Voice preview works but radio is silent? Check microphone, PTT and OpenRouter separately. Sara missing? Select her voice, generate the pack and restart while stopped only after success. 'No data'? Check the source, sim adapter and session state first; a larger model cannot supply missing telemetry. Do not change settings while driving.

12 | First drive, updates and honest limitations

Setup is complete when each stage works separately, before you enter a race.

Before driving

1. Correct RRCA version and sim selected; original installation kept separate.
2. All three voice previews work; Sara is ready and her local pack is loaded.
3. Push-to-talk captures speech and a short test receives an answer.
4. Valid current sim data is visible. Enable the recorder if wanted.
5. Start with a short practice/qualifying/race test. Note the time and unusual calls.

How to handle updates

You can update normal Crew Chief in its separate installation. Replace RRCA with a tested RRCA update; do not install the original Crew Chief installer over the RRCA folder. Public mode blocks that binary updater.

Extract a new ZIP into a new RRCA folder. Keys, memory and profiles are stored outside the ZIP and are kept.

Limitations: no error-free guarantee, no real-world safety device and no complete support for every API field in every sim. Driving tests with more sims are still to come. Missing facts are not invented.

Support: report version/commit, sim, track, session, exact time, expected/heard call and a sanitised log excerpt to the RRCA project. Never upload keys or voice caches.

Official references: OpenRouter Quickstart | ElevenLabs models | OpenRouter privacy | ElevenLabs terms
Project: github.com/JoshuaGerke/RobertsCrewChiefAdvanced. Open rights/trademark questions: FREEWARE_PRUEFUNG.md in the package.